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Lithography

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To the technologists who have made EUVL a reality

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Preface

Extreme ultraviolet lithography (EUVL) is the leading patterning technology for the production of computer chips beyond the current 193-nm-based optical lithography. EUVL has been in development since the 1980s; today, alpha-level EUVL scanners are in the field, and beta-level scanners are expected to be deployed in a few years. EUVL uses photons of 13.5-nm wavelengths for patterning. Because photons are absorbed heavily by all materials at this wavelength, the entire EUVL scanner is in vacuum, and reflective optics—mostly multilayer (ML) mirrors—are used. To develop EUVL as a patterning technology for high-volume manufacturing (HVM) of computer chips, progress has been made on several fronts: EUV light sources, optics, optics metrology, contamination control, masks and mask handling, and resists. In addition, learning from microexposure tools (METs) has been invaluable in the development of EUV resists and full-field EUVL scanners.

This reference book contains 12 chapters contributed by leading researchers and suppliers in the EUVL field. It also contains a detailed appendix with EUVL reference data. The chapter topics are intended to cover the needs of practitioners of the technology as well as readers who want an introduction to EUVL.

This book starts with two chapters on the history of EUVL development. The first chapter covers pioneering work from the earliest days until the EUV Limited Liability Corporation (LLC) was formed to consider the commercial feasibility of EUVL. The contributions of the EUV LLC are covered in the second chapter. Chapter 3 presents an overview of EUV source technology (requirements, technology status, and metrology). Chapters 4a–4d cover EUV optics (substrate, ML deposition, and optical design). Chapter 5 describes and compares different EUV wavefront measurement techniques for EUV optics testing. Chapters 6a–6c explain contamination and its control in EUVL scanners. In today's EUVL scanners, the collector optics in EUV sources are subject to the most serious contamination. Chapter 6 addresses this issue as well as the contamination of scanner optics. Chapter 7 discusses EUV mask technology (substrate, blank fabrication, absorber stack and backside conductive coating, patterning, cleaning, and phase shift masks). Chapter 8 covers the fundamentals and development of EUV resist technology and addresses line edge roughness (LER). Chapter 9 describes the design and components of the first METs, which have enabled resist development. Two of these METs are in the field today. Chapter 10 presents the fundamental design considerations for an EUVL scanner and describes various components of

a full-field scanner, including illumination and projection optics, stages, sensors, and handling systems. Chapter 11 covers the benefits of patterning with 13.5 nm and presents the results of patterning with MET optics; LER and flare in EUVL systems are also discussed in this chapter. Finally, Chapter 12 explains the cost of ownership (CoO) fundamentals and lithography cost trends for future lithography techniques. The appendix contains reference data for the EUV spectral region, information relevant for practitioners of EUVL.

As EUVL technology has continued to develop, I have worked with authors and referees to identify topics that could be covered by a reference book. For detailed information on EUV source technology, readers are encouraged to consult the previously published text, *EUV Sources for Lithography*. Also, because EUVL technology continually evolves, an overview of its status has been omitted from this volume. I expect many readers will be interested in one particular topic and thus go straight to that chapter; for this reason, we have tried to make each chapter as complete in itself as possible.

Over the last two decades, researchers, commercial suppliers, and end users around the world have published several hundred papers per year on EUVL, and the amount of technical data on EUVL technology just continues to increase. My effort as volume editor has been to produce an authoritative reference book on EUVL technology, which until now has not existed. Today, EUV source technology is the leading challenge for implementing EUVL in HVM. EUV mask technology has made significant progress; however, defect-free masks that satisfy defectivity requirements remains a challenge. EUV resist, which will simultaneously satisfy the requirements of resolution, sensitivity, and LER, is another leading challenge for EUVL. As we see continued progress in addressing these three critical technical challenges, through publications and technical conferences, this book will continue to provide the in-depth technical background information relevant to source, mask, resist, scanners, and other technology areas.

The primary strength of this book is that the contributions come from leading experts in each technical area. My initial call for contributions went to authors around the world (Asia, Europe, and the U.S.). Due mostly to time constraints, more U.S. authors are represented in this volume. I am happy to report, however, that each chapter was still reviewed by experts worldwide, who adjusted their busy schedules to review the work of their colleagues and suggest many changes. In this sense, the entire global EUVL community has participated in the generation of this volume. This project has been successful because of the authors' dedication and hard work. This book is the fruit of their labor. During the editing of this volume, I realized that EUVL is a reality today due to the labor of hundreds of technologists around the world. For this reason, I have dedicated this book to these technologists who have made EUVL a reality.

I would also like to acknowledge the contributions of my family, whose influence, encouragement, and support have allowed me to undertake such a project. First of all, my father, Mr. Om Prakash Bakshi, MA, set a very high standard for written communication and the pursuit of excellence, which still today I can only

strive to meet. My mother, Mrs. Pushpa Bakshi, MA, retired lecturer of the Punjabi language, always set an example of hard work and taught me a pragmatic approach toward solving everyday problems, which still guides me. Without the support of my family, Laura and Emily, I would not have been able to complete this project. I very much appreciate their tolerance of my absences in the evenings, on week-ends, and at vacation time, while I labored to complete this volume.

Finally, I would like to thank SPIE Press manager, Timothy Lamkins; SPIE editor, Scott Schrum; and copyeditor, Margaret Thayer. Their teamwork made this project possible. I very much appreciate their support and hard work for making this book project a reality.

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Introduction

The semiconductor industry continues to derive profit and revenue from device scaling, so any technique that promises to extend dimensional scaling receives great attention from the industry. Each new technology node means a lower cost per device and tens of billions of dollars in additional revenue to the industry. As current techniques of extending optical lithography become ever more difficult and costly, each opportunity to introduce a new lithography method opens the door for further cost-effective dimensional scaling.

However, introducing nonoptical lithography techniques requires coordination and resources that no one company can muster on its own. New infrastructure must be created, new supply chains must be established, and new standards must link all of these together. The cost of establishing the infrastructure and developing the tools and processes is usually measured in billions of dollars. Only through the coordination of multiple stakeholders can these dramatic changes occur in a timely manner. Each component of the lithography solution will only be profitable if all of the necessary components are in place. The perfect tool is useless without the right light source, masks, and resists; similarly, the mask, resist, and light source suppliers will not be successful without each other. Risk must be taken, and shared, among all of the parties. In particular, the semiconductor companies will not be successful unless all of the components are ready and manufacturable.

Extreme ultraviolet lithography (EUVL) is one of those non-optical methods that promises extendibility by addressing not only the 32-nm half-pitch nodes, but several nodes beyond that. EUVL provides a photon-based, single-exposure technique that resembles current optical lithography techniques. With a $14\times$ reduction in wavelength compared to 193-nm optical lithography, EUVL provides a significant advantage in diffraction-limited resolution. To ensure a cost-effective, manufacturable solution for EUVL, performance targets have been set that can deliver a very cost-effective, attractive lithography solution.

However, the very short wavelength of EUVL that provides tremendous resolution scalability also hinders the readiness of EUVL technology. The many challenges of using EUV photons to manufacture wafer lithography are reflected in the current EUVL technology challenges. Generating EUV photons from hot and somewhat dirty plasma sources requires new developments in materials for long lifetime and new optics for efficient collection of the clean in-band EUV light. The need for reflective optics places a major burden on the mask blank industry

by requiring the defect-free generation of cost-effective, multilayer-coated mask blanks. Low-projection-optics aberrations for EUV light require polishing to small fractions of the 13.5-nm exposure wavelength. And the high exposure energy can make the generation of low line edge roughness patterns at small dimensions and at low-exposure doses very difficult. These are the challenges and solutions that are discussed in this volume.

As a global consortium of leading-edge semiconductor companies, SEMATECH has been coordinating and leading the EUVL development effort for more than five years. Bringing together semiconductor companies and suppliers through research projects, technical workshops, and broadly scoped lithography symposia has helped catalyze the development of technical results that could make EUVL a manufacturable technology. This volume represents SEMATECH's consensus-building efforts and includes contributions from researchers worldwide. Achieving success in EUVL is truly a global effort, as demonstrated in this volume. Only if we work together can we bring together all of the pieces needed to make EUVL a cost-effective, manufacturable technology.

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List of Abbreviations

AES	Auger electron spectroscopy
AFM	atomic force microscopy
ALD	atomic layer deposition
ALS	Advanced Light Source
AMHS	automated material handling systems
APC	automated process control
APSM	alternating phase shift mask
AR	action required
ARC	antireflective coating
ASET	Association for Super-Advanced Electronics Technologies
ASIC	application specific integrated circuit
BARC	bottom antireflective coating
BCA	binary collision approximation
BIN	COG binary chrome on glass
BNL	Brookhaven National Laboratory
BW	bandwidth
CA	clear aperture
CAR	chemically amplified photoresist
CCD	charge-coupled device
CCOS	computer-controlled surfacing
CD	critical dimension
CE	conversion efficiency
CGH	computer-generated hologram
CMM	coordinate measuring machine
CMP	chemical-mechanical polishing
CoO	cost of ownership
CP	cell projection
CPMI	Center for Plasma Material Interactions
CRADA	cooperative research and development agreement
CTE	coefficient of thermal expansion
CTF	contrast transfer function
CVD	chemical vapor deposition
CW	continuous wave
CXRO	Center for X-ray Optics

DARPA	Defense Advanced Research Projects Agency
dc	direct current
DDR&E	Director of Defense Research & Engineering
DLS	damped least squares
DMAP-MDS	dimethylaminopentamethyldisilane
DMT	debris mitigation tool
DOC	Department of Commerce
DOD	Department of Defense
DOE	Department of Energy
DOE	design of experiments
DOF	depth of focus
DP	double patterning
DPF	dense plasma focus
DPP	discharge-produced plasma
DRAM	dynamic random access memory
DSP	digital signal processor
DUV	deep ultraviolet
E-D	exposure-defocus
EIPBN	Electron, Ion, and Photon Beam Technology and Nanofabrication
EIT	extreme ultraviolet imaging telescope
EL	exposure latitude
EPL	electron-beam projection lithography
ePSM	embedded phase shift mask
ESA	energy sector analyzer
ESCAP	environmentally stable chemically amplified photoresists
ESL	etch stop layer
ETS	Engineering Test Stand
EUCLIDES	Extreme UV Concept Lithography Development System
EUV	extreme ultraviolet
EUVA	Extreme Ultraviolet Lithography System Development Association
EUVL	extreme ultraviolet lithography
EUV LLC	EUV Limited Liability Corporation
EXTATIC	Extreme UV Alpha Tools Integration Consortium (MEDEA+)
FC	Flying Circus
FEA	finite element analysis
FEL	free electron laser
FEM	focus exposure matrix
FF	far field
FIA	field image alignment
FAB	fabrication facility
FIB	focus ion beam
FOM	Fundamenteel Onderzoek der Materie Institute
FOUP	front-opening unified pod

FTE	full-time equivalent
FTIR	Fourier transform infrared spectroscopy
FVC	flare variation compensation
FWHM	full-width half maximum
GAE	gas-assisted etching
GI	grazing incidence
GUI	graphical user interface
GWLE	good wafer level exposure
H	horizontal
HIT	Himeji Institute of Technology
HMDS	hexamethyldisilazane
HP	half-pitch
HPFS	high-purity fused silica
HSFR	high-spatial-frequency roughness
HSQ	hydrogen silsesquioxane
HV	horizontal/vertical
HVM	high-volume manufacturing
HWS	Hartmann wavefront sensor
IBSD	ion beam sputter deposition
IBA	inverse bremsstrahlung absorption
IBF	ion beam figuring
IL	imaging layer
IC	integrated circuit
ICA	instantaneous clear aperture
IF	intermediate focus
ILS	image log slope
INERT	Illinois ion energy reduction technique
IP	intellectual property
IPD	in-plane distortion
IPE	image placement error
IPL	ion-beam projection lithography
IPS	integrated product scheduling
ISMI	International SEMATECH Manufacturing Initiative
ISMT	International SEMATECH
ITRS	International Technology Roadmap for Semiconductors
JDA	joint development agreement
JDP	joint development program
JSPE	Japan Society for Precision Engineering
LANL	Los Alamos National Laboratory
LBNL	Lawrence Berkeley National Laboratory
LDD	low defect deposition
LiEDA	lithium salt of ethylenediamine
LER	line edge roughness
LLNL	Lawrence Livermore National Laboratory

LOCOS	local oxidation of silicon
LPCVD	low-pressure chemical vapor deposition
LPP	laser-produced plasma
L/S	line/space
LSC	laser shock wave cleaning
LSI	large-scale integration
LSM	layered synthetic microstructure
LTEM	low thermal expansion material
LTF	line edge roughness transfer function
LTO	low-temperature oxide
LTR	LER transfer function
LWR	line width roughness
MBDC	Mask Blank Development Center (SEMATECH)
MBE	molecular beam epitaxy
MCP	microchannel plate
MEDEA+	Microelectronics Development for European Applications+
MEEF	mask error enhancement factor
MET	microexposure tool
MFS	minimum feature size
MHD	magnetohydrodynamics
ML	multilayer
ML2	maskless lithography
MLM	multilayer mirror
MOPA	master oscillator–power amplifier
MOS	metal-oxide semiconductor
MPU	microprocessing unit
MRF	magnetorheological finishing
MSFR	mid-spatial-frequency roughness
MTF	modulation transfer function
MTBF	mean time between failure
MTTR	mean time to repair
MTTT	mean time to test
Mw	molecular weight
NA	numerical aperture
NGL	next-generation lithography
NI	normal incidence
NIL	nano imprint lithography
NILS	normalized image log slope
NIST	National Institute of Standards and Technology
NMOS	n-type metal-oxide semiconductor
NSLS	National Synchrotron Light Source
NSR	Nikon Step and Repeat System
OAI	off-axis illumination
OH	overhead

OL	overlay
OOB	out-of-band
OPC	optical proximity correction
OPD	out-of-plane distortion
OSA	Optical Society of America
OTF	optical transfer function
PACE	plasma-assisted cleaning by electrostatics
PAG	photo acid generator
PBS	polybutene-1 sulfone
PDI	point-diffraction interferometry
PEB	post-exposure bake
PECVD	plasma-enhanced chemical vapor deposition
PEEM	photoelectron emission microscopy
PHS	polyhydroxystyrene
PM	preventative maintenance
PMI	phase-measuring interferometer
PMM	phase-measuring microscopy
PMMA	poly (methyl methacrylate)
PO	projection optics
POB	projection optics box
PPL	planarization layer
PREUVE	PRogramme Extreme UV
PSI	Paul Scherrer Institut
PS/PDI	phase-shifting point-diffraction interferometer
PSD	power spectral density
PSDI	phase-shifting diffraction interferometer
PSF	point-spread function
PSL	polystyrene latex
PSM	phase shift mask
PSPDI	phase-shifting point-diffraction interferometer
PTB	Physikalisch-Technische Bundesanstalt
PV	peak to valley
PVD	physical vapor deposition
QCM	quartz crystal microbalance
RAM	reliability, availability, and maintainability
RDC	Research Development Center (of the VNL)
RES	resolution
RET	resolution enhancement technique
RGA	residual gas analyzer
RH	relative humidity
RIM	reticle imaging microscope
RIE	reactive ion etch
RLS	resolution, LER, sensitivity
RM	reflection mask

ROFR	right of first refusal
ROI	return on investment
ROMP	ring-opening metathesis polymerization
RP1	Ronler Processing 1
SAHD	send-ahead
SB	soft bake
SEM	scanning electron microscopy
SEMI	Semiconductor Equipment and Materials International
SES	Subfield Exposure Station
SHC	supersonic hydrocleaning
SHWS	Shack-Hartmann wavefront sensor
SIA	Semiconductor Industry Association
SIMS	secondary-ion mass spectroscopy
SLR	single-layer resist
SMIF	standard mechanical interface
SNL	Sandia National Laboratories
SOHO	solar and heliospheric observatory
SOW	statements of work
SPC	statistical process control
SPF	spectral purity filter
SPP	silicone-based positive photoresist
SRAF	sub-resolution assist features
SRC	Semiconductor Research Corporation
STM	scanning tunneling microscopy
SVGL	Silicon Valley Group Lithography
SXPL	soft x-ray projection lithography
TEG	test elements group
TEM	transmission electron microscope
TIS	total integrated scatter
TM	transmission mask
TMAH	tetramethylammonium hydroxide
TPT	throughput
TRACE	transition region and coronal explorer
TSI	top surface imaging
UHV	ultrahigh vacuum
ULE	ultra-low-expansion (Corning glass)
UPW	ultra-pure water
USAL	United States Advanced Lithography
UFTL	unexposed film thickness loss
UTR	ultrathin resist
VNL	Virtual National Laboratory
VUV	vacuum ultraviolet
VSF	vector-shaped beam
WPM	wafers per mask

WFE	wavefront error
WPH	wafers per hour
XPL	x-ray proximity lithography
XPS	x-ray photoelectron spectroscopy
XRD	x-ray diffraction
YAG	yttrium aluminum garnet